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Stochastic Dynamic Game Nash Seeking Equilibrium

Problem Formulation

Consider a network of N agents, where each agent $i \in 1, \dots, N$ is described by the linear stochastic dynamics:

$$\dot{x}_i(t) = Ax_i(t) + Bu_i(t) + w_i(t), \quad (1)$$

where $x_i(t) \in \mathbb{R}^d$ is the state, $u_i(t) \in \mathbb{R}^m$ is the control input, and $w_i(t)$ is a stochastic disturbance.

The communication between agents is described by an undirected connected graph $\mathcal{G}$ with Laplacian matrix L .

Cost Function and Game Formulation

Each agent minimizes the following cost functional:

$$J_i = \mathbb{E} \left[\int_0^T \left(\sum_{j \in \mathcal{N}_i} |x_i - x_j|^2 + |u_i|^2 \right) dt \right]. \quad (2)$$

This defines a non-cooperative dynamic game, where each agent seeks to minimize its own cost while being coupled with the others. The control effort quantifies the amount of energy expended by the agents in order to achieve consensus. In the proposed framework, the control effort is represented by the quadratic term

$$|u_i(t)|^2,$$

which penalizes excessively large control inputs. The total control effort over the simulation horizon is defined as

$$U_i = \int_0^T |u_i(t)|^2 dt.$$

This term plays a crucial role in balancing convergence performance and energy consumption. Large control inputs may accelerate convergence, but they also increase the overall energy expenditure. Consequently, the weighting matrix (R) in the LQR formulation determines the trade-off between fast consensus and smooth control action.

Nash Equilibrium

A set of strategies $u_{i=1}^{*N}$ is a Nash equilibrium if:

$$J_i(u_i^*, u_{-i}^*) \leq J_i(u_i, u_{-i}^*), \quad \forall u_i. \quad (3)$$

Derivation of the Nash Strategy

We consider a gradient-based approach. The instantaneous cost is:

$$\ell_i(x, u_i) = \sum_{j \in \mathcal{N}_i} |x_i - x_j|^2 + |u_i|^2. \quad (4)$$

The first term represents the disagreement between neighboring agents, whereas the second term penalizes excessive control effort.

In order to derive the Nash strategy, we examine the direction in which the disagreement cost increases most rapidly. For this reason, we compute the gradient of the disagreement term:

$$\|x_i - x_j\|^2 = (x_i - x_j)^T (x_i - x_j)$$

Expanding the quadratic term gives

$$\|x_i - x_j\|^2 = x_i^T x_i - 2x_i^T x_j + x_j^T x_j$$

Taking the gradient with respect to x_i :

$$\nabla_{x_i} \ell_i = 2 \sum_{j \in \mathcal{N}_i} (x_i - x_j). \quad (5)$$

The gradient vector indicates the direction of maximum increase of the disagreement cost. Consequently, in order to reduce disagreement among neighboring agents, the control input must act in the opposite direction of the gradient. We propose the distributed feedback strategy:

$$u_i = -K \sum_{j \in \mathcal{N}_i} (x_i - x_j), \quad (6)$$

where $K \in \mathbb{R}^{m \times d}$ is a feedback gain matrix. The proposed control law is fully distributed since each agent requires only relative information from its neighboring agents rather than global information from the entire network.

Closed-Loop Dynamics

Substituting the control law into the dynamics yields:

$$\dot{x}_i = Ax_i - BK \sum_{j \in \mathcal{N}_i} (x_i - x_j) + w_i. \quad (7)$$

The term Ax_i represents the intrinsic dynamics of each agent, whereas the coupling term

$$BK \sum_{j \in \mathcal{N}_i} (x_i - x_j)$$

drives the agents toward consensus.

In compact global form, the overall network dynamics can be written as

$$\dot{x} = (I_N \otimes A - L \otimes BK)x + w. \quad (8)$$

where $I_N \otimes A$ describes the intrinsic dynamics of all agents, $L \otimes BK$ captures the coupling structure induced by the communication graph.

Consensus Nash Equilibrium

Proposition 1: The Nash equilibrium of the game corresponds to a consensus configuration.

Proof:

At equilibrium, $\dot{x}_i = 0$ and $w_i = 0$, hence:

$$Ax_i - BK \sum_{j \in \mathcal{N}_i} (x_i - x_j) = 0. \quad (9)$$

If all agents satisfy $x_i = x_j = x^*$, then:

$$\sum_{j \in \mathcal{N}_i} (x_i - x_j) = 0, \quad (10)$$

and thus:

$$Ax^* = 0. \quad (11)$$

and therefore

$$u_i = 0$$

which implies that no additional control effort is required once consensus has been achieved.

Furthermore, any unilateral deviation from the consensus point increases the disagreement term $\|x_i - x_j\|^2$ and consequently increases the local cost function. Therefore, no agent has an incentive to independently deviate from the equilibrium configuration, implying that the consensus point constitutes a Nash equilibrium.

For the stability analysis, the spectral decomposition of the Laplacian matrix is employed:

$$L = U\Lambda U^T$$

where $\Lambda = \text{diag}\{\lambda_1, \dots, \lambda_N\}$. contains the eigenvalues of the graph Laplacian. Since the communication graph is connected,

$$\lambda_1 = 0, \lambda_i > 0, i \geq 2$$

Applying modal decomposition, the closed-loop system decouples into independent modes:

$$\dot{z}_i = (A - \lambda_i BK)z_i$$

The mode associated with

$$\lambda_1 = 0$$

corresponds to the consensus subspace, whereas the remaining modes describe the disagreement dynamics. Consensus is achieved if all disagreement modes are asymptotically stable, namely if the matrices $A - \lambda_i BK$ are Hurwitz for every $\lambda_i > 0$. Under this condition,

$$z_i(t) \rightarrow 0$$

which directly implies

$$x_i(t) - x_j(t) \rightarrow 0$$

thereby proving asymptotic consensus among all agents. If A is stable or has equilibrium at the origin, then $x^* = 0$ is a consensus Nash equilibrium. ■

Stability Analysis

Proposition 2: The closed-loop system achieves consensus if the matrices $A - \lambda_i BK$ are Hurwitz for all nonzero eigenvalues λ_i of L .

Proof:

Let $x = (\mathbf{1} \otimes x_c) + \tilde{x}$, where $\tilde{x}$ lies in the disagreement subspace.

Using the spectral decomposition of L , the disagreement dynamics become:

$$\dot{\tilde{x}} = (I \otimes A - L \otimes BK)\tilde{x}. \quad (12)$$

This decomposes into modes:

$$\dot{z}_i = (A - \lambda_i BK)z_i. \quad (13)$$

If all matrices $A - \lambda_i BK$ are Hurwitz for $\lambda_i > 0$, then all disagreement modes converge to zero, implying consensus. ■

Lyapunov Stability

Consider the Lyapunov function:

$$V = \frac{1}{2}x^T(L \otimes I_d)x, \quad (14)$$

where L is the Laplacian matrix of the communication graph and I_d is the identity matrix.

Since L is symmetric for an undirected graph, we have:

$$\dot{V} = x^T(L \otimes I_d)\dot{x}. \quad (15)$$

Consider the closed-loop dynamics (without noise):

$$\dot{x} = (I \otimes A - L \otimes BK)x. \quad (16)$$

Substituting into the Lyapunov derivative yields:

$$\dot{V} = x^T(L \otimes I_d)(I \otimes A - L \otimes BK)x \quad (17)$$

$$= x^T [(L \otimes A) - (L^2 \otimes BK)] x. \quad (18)$$

Using the eigen-decomposition of the Laplacian $L = U\Lambda U^T$, define the transformed state $\tilde{x} = (U^T \otimes I_d)x$. Then the dynamics decouple into modes:

$$\dot{\tilde{x}}_i = (A - \lambda_i BK)\tilde{x}_i, \quad (19)$$

where λ_i are the eigenvalues of L .

Therefore, the Lyapunov derivative can be written as:

$$\dot{V} = \sum_{i=1}^N \tilde{x}_i^T (A - \lambda_i BK)\tilde{x}_i. \quad (20)$$

Under appropriate gain selection such that $A - \lambda_i BK$ is Hurwitz for all non-zero eigenvalues λ_i , it follows that:

$$\dot{V} \leq 0, \quad (21)$$

which guarantees convergence to the consensus subspace.

Stochastic Extension

With noise:

$$\dot{x} = (I \otimes A - L \otimes BK)x + w, \quad (22)$$

the system becomes a stochastic differential equation, where w represents an external stochastic disturbance. Under this formulation, the system can be interpreted as a stochastic differential equation or a noise-perturbed deterministic system, depending on the properties of w . The system achieves mean-square consensus:

$$\lim_{t \rightarrow \infty} \mathbb{E}|x_i(t) - x_j(t)|^2 = 0. \quad (23)$$

In the presence of persistent stochastic noise, the system achieves mean-square bounded consensus, i.e.,

$$\limsup_{t \rightarrow \infty} \mathbb{E}\|x_i(t) - x_j(t)\|^2 \leq \epsilon_w \quad (24)$$

where $\epsilon_w > 0$ depends on the noise intensity and the network/control gains.

However, due to the presence of persistent noise, the agent states do not converge exactly to the equilibrium point, but instead fluctuate around it within a bounded neighborhood.

Consider the closed-loop multi-agent system in the presence of additive stochastic disturbances:

$$dx = \left((I_N \otimes A - L \otimes BK)x \right) dt + (I_N \otimes G)dW_t, \quad (25)$$

where:

- $x \in \mathbb{R}^{Nn}$ is the stacked state vector,
- L is the graph Laplacian,
- A, B, K are the system and controller matrices,
- G is the noise intensity matrix,
- W_t is a standard Wiener process.

The stochastic disturbance transforms the deterministic closed-loop dynamics into an Itô stochastic differential equation.

Theorem 0.1 (Mean-Square Bounded Consensus). Assume that:

1. The communication graph is connected,
2. The feedback gain K is selected such that

$$A - \lambda_i BK$$

is Hurwitz for every nonzero Laplacian eigenvalue λ_i , $i = 2, \dots, N$,

3. The disturbance covariance is bounded.

Then, the stochastic multi-agent system (25) achieves mean-square bounded consensus, namely,

$$\limsup_{t \rightarrow \infty} \mathbb{E}\|x_i(t) - x_j(t)\|^2 \leq \epsilon_w, \quad (26)$$

for all $i, j \in \{1, \dots, N\}$, where $\epsilon_w > 0$ depends on the disturbance intensity and the closed-loop damping.

Proof. Define the disagreement vector

$$\xi = (M \otimes I_n)x, \quad (27)$$

where

$$M = I_N - \frac{1}{N}\mathbf{1}\mathbf{1}^T$$

is the disagreement projection matrix satisfying

$$M\mathbf{1} = 0.$$

Applying the transformation to (25) yields

$$d\xi = \left((I_N \otimes A - L \otimes BK)\xi \right) dt + (M \otimes G)dW_t. \quad (28)$$

Consider the quadratic Lyapunov function

$$V(\xi) = \xi^T (P \otimes I_n) \xi, \quad (29)$$

where $P = P^T > 0$.

Using Itô's formula,

$$\begin{aligned} dV = & \left[\xi^T \left(\Phi^T P + P \Phi \right) \xi + \text{Tr} \left((M \otimes G)^T P (M \otimes G) \right) \right] dt \\ & + 2\xi^T P (M \otimes G) dW_t, \end{aligned} \quad (30)$$

where

$$\Phi = I_N \otimes A - L \otimes BK.$$

Taking mathematical expectation eliminates the martingale term:

$$\begin{aligned} \frac{d}{dt} \mathbb{E}[V] = & \mathbb{E} \left[\xi^T (\Phi^T P + P \Phi) \xi \right] \\ & + \text{Tr} \left((M \otimes G)^T P (M \otimes G) \right). \end{aligned} \quad (31)$$

Since the graph is connected and the matrices

$$A - \lambda_i BK$$

are Hurwitz for all nonzero λ_i , there exists $Q = Q^T > 0$ such that

$$\Phi^T P + P \Phi = -Q. \quad (32)$$

Substituting into (31) gives

$$\frac{d}{dt} \mathbb{E}[V] \leq -\lambda_{\min}(Q) \mathbb{E} \|\xi\|^2 + c_w, \quad (33)$$

where

$$c_w = \text{Tr} \left((M \otimes G)^T P (M \otimes G) \right).$$

Using standard comparison arguments,

$$\limsup_{t \rightarrow \infty} \mathbb{E} \|\xi(t)\|^2 \leq \frac{c_w}{\lambda_{\min}(Q)}. \quad (34)$$

Since ξ represents the disagreement dynamics, there exists a constant $\epsilon_w > 0$ such that

$$\limsup_{t \rightarrow \infty} \mathbb{E} \|x_i(t) - x_j(t)\|^2 \leq \epsilon_w.$$

Therefore, the agents achieve mean-square bounded consensus despite persistent stochastic disturbances. $\square$

The above result shows that stochastic disturbances prevent exact asymptotic consensus, but the disagreement remains bounded in the mean-square sense. The residual consensus error depends on the disturbance covariance and the closed-loop stability margin.

LQR-Based Design of the Feedback Gain

In this section, we derive the feedback gain K using a Linear Quadratic Regulator (LQR) approach, ensuring optimality and closed-loop stability.

Aggregate System Formulation

Consider the stacked state vector:

$$x = [x_1^T, \dots, x_N^T]^T \in \mathbb{R}^{Nd}. \quad (35)$$

The global dynamics can be written as:

$$\dot{x} = (I_N \otimes A)x + (I_N \otimes B)u + w. \quad (36)$$

Define the aggregate cost:

$$J = \mathbb{E} \left[\int_0^T (x^T (L \otimes Q)x + u^T (I_N \otimes R)u) dt \right], \quad (37)$$

where $Q \succeq 0$ and $R \succ 0$ are weighting matrices.

Interpretation

The term:

$$x^T (L \otimes Q)x = \sum_{i,j} a_{ij} |x_i - x_j|_Q^2$$

penalizes disagreement between agents, enforcing consensus, while the control penalty ensures bounded inputs.

Optimal Control Law

The problem is a standard LQR problem for the system:

$$\dot{x} = \mathcal{A}x + \mathcal{B}u,$$

with:

$$\mathcal{A} = I_N \otimes A, \quad \mathcal{B} = I_N \otimes B.$$

The optimal control law is:

$$u^* = -\mathcal{K}x, \quad (38)$$

where:

$$\mathcal{K} = (I_N \otimes R^{-1}B^T)P, \quad (39)$$

and P is the unique positive semidefinite solution of the Algebraic Riccati Equation (ARE):

$$\mathcal{A}^T P + P\mathcal{A} - P\mathcal{B}(I_N \otimes R^{-1})\mathcal{B}^T P + (L \otimes Q) = 0. \quad (40)$$

Decentralized Structure

Due to the Kronecker structure, the solution admits a distributed implementation. In particular, the optimal feedback can be expressed as:

$$u_i = -K \sum_{j \in \mathcal{N}_i} (x_i - x_j), \quad (41)$$

where:

$$K = R^{-1} B^T P_0, \quad (42)$$

and P_0 is the solution of the reduced Riccati equation:

$$A^T P_0 + P_0 A - P_0 B R^{-1} B^T P_0 + Q = 0. \quad (43)$$

Connection to Nash Equilibrium

The LQR-based feedback gain yields a control law that minimizes a global quadratic cost. Due to the potential structure of the game, this solution coincides with a Nash equilibrium of the original distributed problem.

Stability Guarantee

The closed-loop dynamics become:

$$\dot{x} = (I_N \otimes A - L \otimes BK)x. \quad (44)$$

Since $P_0 \succ 0$ solves the Riccati equation, the matrix:

$$A - \lambda_i BK$$

is Hurwitz for all nonzero eigenvalues λ_i of L .

Therefore, all disagreement modes are asymptotically stable, and the system achieves consensus.

Discussion

The LQR-based design provides a systematic way to compute the feedback gain K , balancing convergence speed and control effort. Moreover, it guarantees stability and optimality, making it suitable for large-scale multi-agent systems.

Discussion

The proposed framework demonstrates that a distributed Nash strategy leads to consensus behavior in multi-agent systems with linear dynamics. The inclusion of stochastic disturbances highlights the robustness of the approach, while the control effort remains bounded due to the quadratic penalty term.

Simulation

The behavior of the proposed stochastic dynamic game is illustrated through numerical simulations.

Figure 1 depicts the trajectories of all agents in the state space. Despite different initial conditions, all agents converge to a common point, confirming the emergence of a consensus Nash equilibrium. The convergence is smooth, indicating that the distributed feedback law successfully stabilizes the multi-agent system.

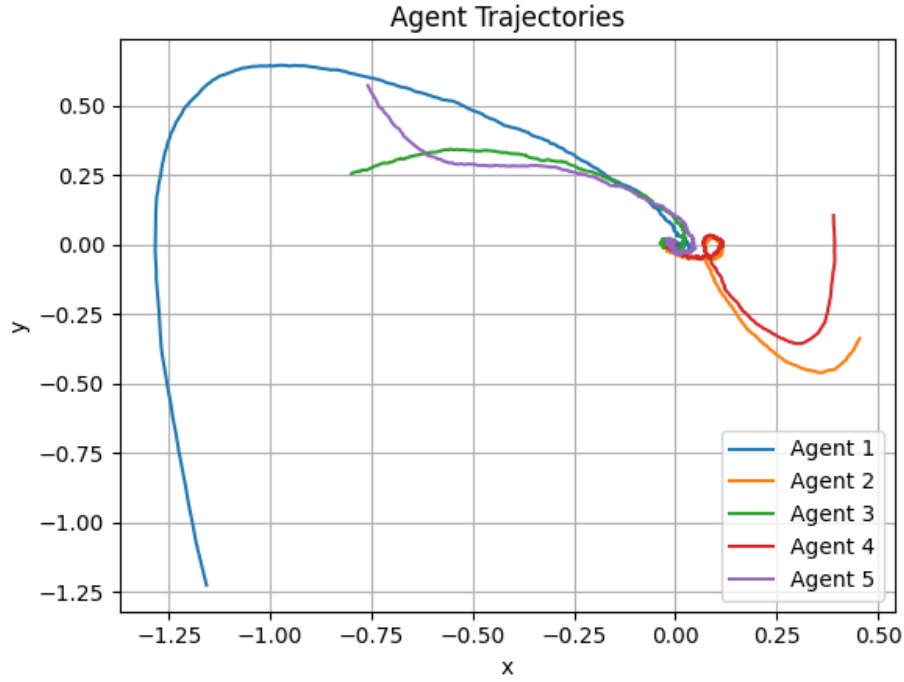


Figure 1: Trajectories of agents

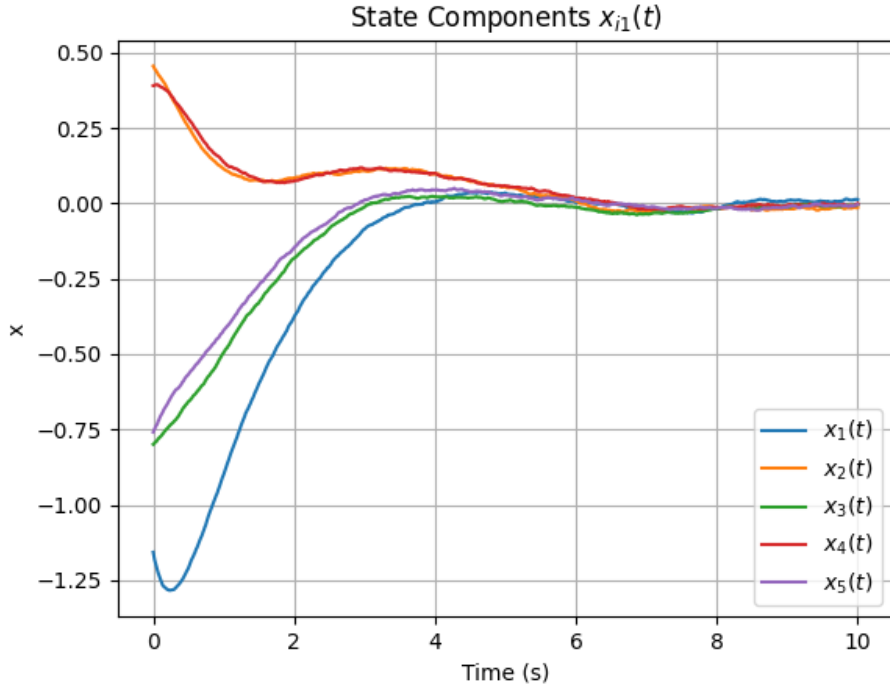


Figure 2: Trajectory x_{i1} with respect to time.

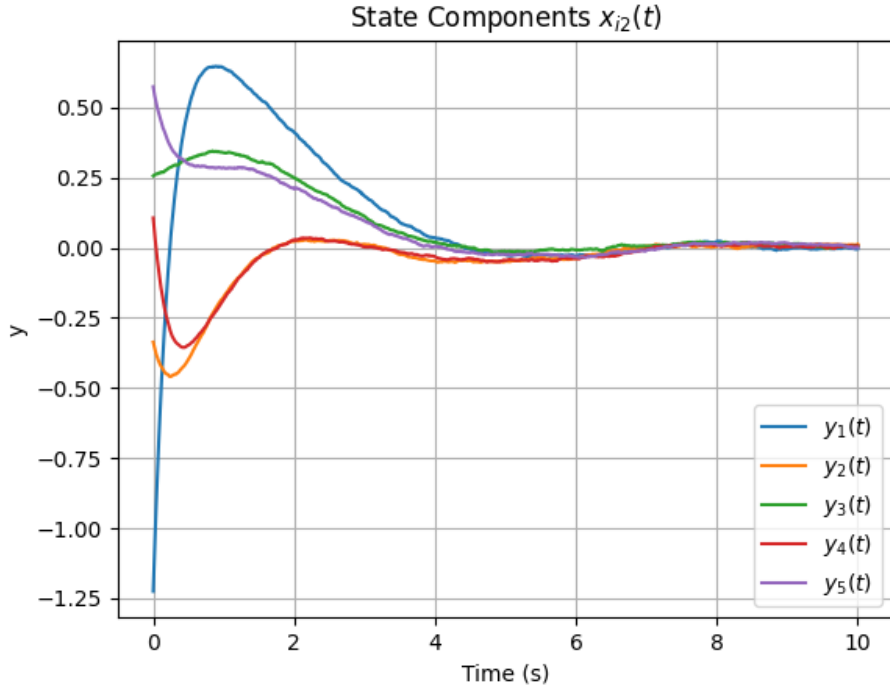


Figure 3: Trajectory x_{i2} with respect to time.

Figures 2 and 3 present the time evolution of the state components $x_{i1}(t)$ and $x_{i2}(t)$, respectively. It can be observed that all state trajectories asymptotically approach each other, demonstrating consensus across all agents. Due to the presence of stochastic disturbances, small fluctuations persist around the equilibrium point. This behavior is consistent with mean-square convergence, rather than exact deterministic convergence.

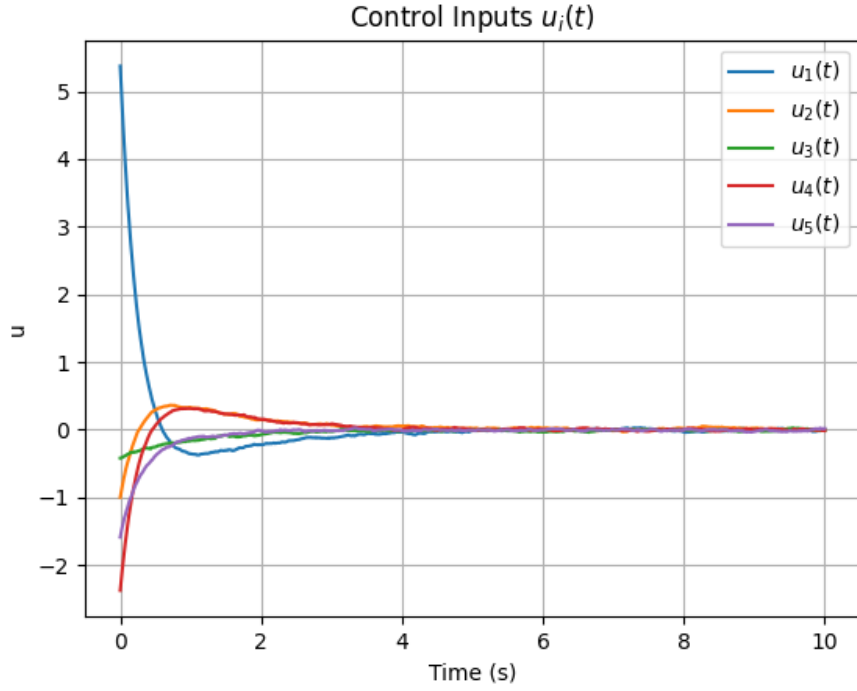


Figure 4: Control inputs $u_i(t)$ with respect to time.

Figure 4 illustrates the control inputs $u_i(t)$ for all agents. Initially, the inputs exhibit relatively large magnitudes, reflecting the significant disagreement among agents. As the system evolves and consensus is achieved, the control inputs decrease and approach zero. The small oscillations observed at steady state are due to the stochastic disturbances affecting the system dynamics.

Overall, the simulation results validate the theoretical analysis, demonstrating that the proposed distributed Nash strategy ensures consensus, stability, and robustness in the presence of noise.

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